LindfoSec VMDR Practice Lab

Phase 1 Summary — Environment Build & Asset Organization

A hands-on vulnerability management practice project built with Qualys VMDR, modeled on an international enterprise (LindfoSec) headquartered in Asturias, Spain. This document summarizes Phase 1 — environment build, asset organization, and Qualys tagging/grouping strategy.

1. Plain-language overview of Phase 1

Phase 1 built a small but realistic practice environment for vulnerability management. The idea: pretend LindfoSec is a real international company with five departments, headquartered in Asturias, Spain. Instead of one messy test server, each department got its own cloud server in AWS, so the lab mirrors what a Vulnerability Management, Detection & Response (VMDR) analyst actually deals with — many machines, different owners, different levels of sensitivity.

In simple terms: five AWS servers were built (one per department), each running a different Linux distribution so the vulnerabilities found would look realistic and varied rather than identical across the board. Each server was deliberately given one known outdated component so that the first Qualys scan would return meaningful, explainable findings instead of a blank report.

The other half of Phase 1 was organizational, not technical: deciding how those five servers should be labeled and grouped inside Qualys so that reporting, prioritization, and access control can be done by department and by location — the same way a real security team would slice up a global estate.

2. Environment summary

Region: eu-south-2 (AWS Europe — Spain), chosen as the LindfoSec HQ region. On-demand pricing for t3.micro is identical to eu-west-1 (Ireland) at $0.0114/hr, so the choice was made for narrative fit (a real Spain-based region) rather than cost.

Department	OS	Instance	Subnet	Seeded weakness
IT / Mgmt	Ubuntu 20.04 LTS	t3.micro	Private	Older OpenSSH / sudo version
Finance	Amazon Linux 2	t3.micro	Private	Outdated OpenSSL, weak TLS config
HR	Ubuntu 20.04 LTS	t3.micro	Private	Older MariaDB/MySQL version
Sales & Marketing	Ubuntu 22.04 LTS	t3.micro	Public	Outdated Apache/PHP or WordPress
R&D / Engineering	Debian 11	t3.micro	Private	Old Node.js/Docker, unpatched dev tools

On-prem endpoint: Windows laptop, representing a corporate/executive device managed alongside the AWS estate via the same Qualys Cloud Agent — demonstrating hybrid (cloud + on-prem) asset management.

3. Asset Groups & Tagging — why it matters

Qualys gives you two overlapping ways to organize assets, and a real analyst needs to know when to use each.

	Asset Groups (legacy)	Asset Tags (modern)
Purpose	Static bucket for scan scoping and user access control	Dynamic/rule-based labeling that drives dashboards, TruRisk, and reporting
Membership	Manually add IPs/ranges	Manual or automatic, via rules (e.g. hostname contains 'fin-')
Best for	Restricting what a user/role can see or scan	Prioritization, filtering, cross-cutting views (e.g. dept + location + criticality)

LindfoSec structure used in the lab

Asset Groups (one per department, static, plus one umbrella group):

- AG-IT-Mgmt, AG-Finance, AG-HR, AG-Sales-Marketing, AG-RnD (one instance IP each)
- AG-HQ-Asturias — all five phase-1 instances (location-based group)
- AG-LindfoSec-All — global umbrella group, used for company-wide reporting

Asset Tags (hierarchical, two independent facets — department and location):

- Dept: IT-Mgmt / Finance / HR / Sales-Marketing / RnD
- Location: HQ-Asturias-Spain (Americas and APAC branch tags added in later phases)
- Criticality: High / Medium / Low
- Exposure: Public / Private

Because tags are not mutually exclusive, an asset carries multiple tags at once. This lets Qualys answer questions Asset Groups alone can't — e.g. “all Finance assets globally” (Dept tag only), “everything in the Americas branch” (Location tag only), or “Critical findings on EU-based assets” (Location + Criticality combined — the GDPR/data-residency angle).

4. Step-by-step: creating an Asset Group in Qualys

1. Log in to the Qualys Cloud Platform and open the VMDR (or Asset Management) module.
2. Go to the Assets tab, then select Asset Groups from the sub-menu.
3. Click New > Asset Group.
4. Enter a clear Title (e.g. AG-Finance) — keep naming consistent across departments.
5. Add the relevant IP addresses / IP ranges, or select existing host assets to include in the group.
6. Optionally assign a Business Unit and set an Asset Group Owner to control who manages it.
7. Click Save.
8. Repeat for each department, then create one umbrella group (e.g. AG-LindfoSec-All) containing every asset, for company-wide reporting.

Use Asset Groups when the goal is scoping — restricting a scan, a report, or a user's visibility to one department at a time.

5. Step-by-step: creating an Asset Tag in Qualys

1. Open the Assets tab and select Tags (available from VMDR or the Asset Inventory / CSAM module).
2. Click New Tag (or the '+' / Create Tag button).
3. Enter a Tag Name. For a hierarchy, create the parent first (e.g. LindfoSec), then create child tags nested underneath it (e.g. Dept: Finance) using the parent-tag selector.
4. Choose the Rule Engine option: select No Dynamic Rule for a simple static tag, or pick a rule type (e.g. Asset Name Contains, Network Range, Cloud Instance Metadata, Instance State) to make the tag dynamic.
5. If dynamic, enter the matching value (e.g. Instance Name Contains = fin-) so any current or future asset matching that pattern is tagged automatically.
6. Click Save. Dynamic tags evaluate immediately against existing assets and continue to apply to new ones; static tags must be applied manually.
7. To manually tag an asset: go to Asset Inventory, select the asset(s), choose Actions > Apply Tags, select the tag, and save.
8. Verify the setup by filtering Asset Inventory by tag — confirm the expected assets appear under each Dept, Location, Criticality, and Exposure tag.

Use Tags when the goal is cross-cutting analysis and prioritization — filtering and reporting across department, location, and risk dimensions at once, which is what modern VMDR dashboards and TruRisk scoring are built around.